PROJECT DOBAARA

TIKAU PHONES, JANUARY TO MARCH 2027

A board recommendation: which options to fund from ₹80 lakh of growth money, which customer to win, how to launch in February and how to know by mid February that it is working

OUR ANSWER: CERTIFY EVERY PHONE, PROMISE 85% BATTERY AND 12 MONTHS, AND OPEN A THIRD LAB SHIFT ON 1 FEBRUARY, FOR FAMILY BUYERS

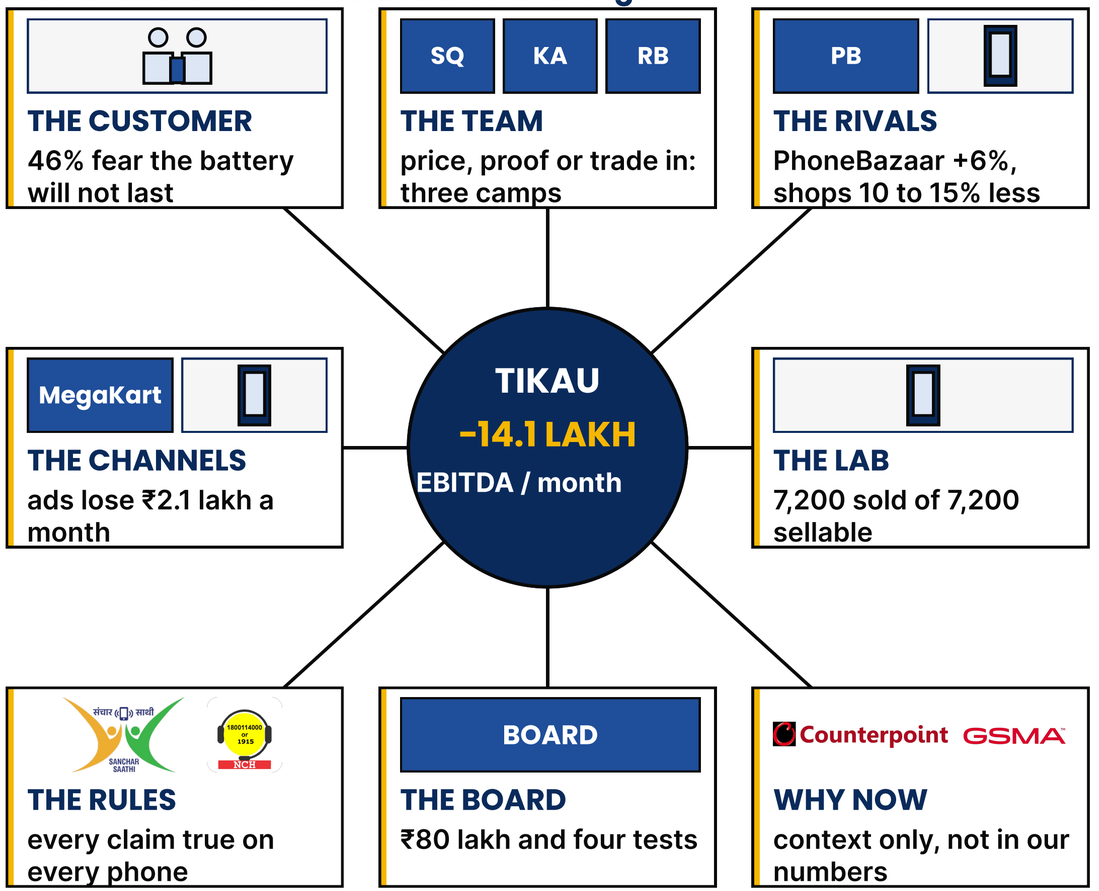
March EBITDA improves from −₹14.1 lakh to +₹5.6 lakh; returns fall to 4.5%, using ₹58.0 lakh. Numerical tests pass; claims require Exhibit 5 controls (A4). Trade in waits for April.

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member
Dayita Arora	Ramjas College, University of Delhi	Member

WHO DECIDES WHETHER A TIKAU PHONE GETS BOUGHT

One Noida lab, five channels, three grades: Tikau sells 7,200 phones a month and loses ₹14.1 lakh a month doing it.



Native illustrations; fictional-name monograms. Official marks identify case bodies.

Source: case Exhibits 1 to 6 | team model 9b3628d94d | logos identify organisations named in the case; no partnership or endorsement is implied

EXECUTIVE SUMMARY

Certify every phone, promise 85% battery and 12 months, and open a third lab shift on 1 February: March EBITDA moves from −₹14.1 lakh to +₹5.6 lakh

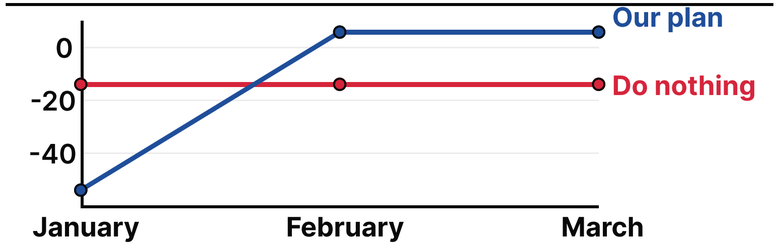
+19.7
₹ lakh a month better in March than doing nothing

58.0 of ₹80
B1 ₹15 + B2 ₹25; shift ₹9 in February and March

4.5% returns
limit 6.0% in March

5 of 84 plans
Conditional screen; highest Q1 EBITDA

MONTHLY EBITDA, ₹ LAKH



Q1: −₹0.5 lakh vs no change; A3.

THE DECISION TABLE

₹ lakh	January	February	March
Options live	Setup only	B1, B2, shift	B1, B2, shift
Phones sold	7,200	8,328	8,328
Contribution	43.9	72.6	72.6
Growth money	40.0	9.0	9.0
EBITDA	−54.1	5.6	5.6
Return rate	8.1%	4.5%	4.5%

THE BOARD'S FOUR TESTS

	Limit	Ours	Result
1 Growth money	₹80 lakh	₹58.0	Pass
2 March EBITDA	₹0 or more	+5.6	Pass
3 March returns	6.0%	4.5%	Pass
4 Exhibit 5 claims	Controls	Required	Pass

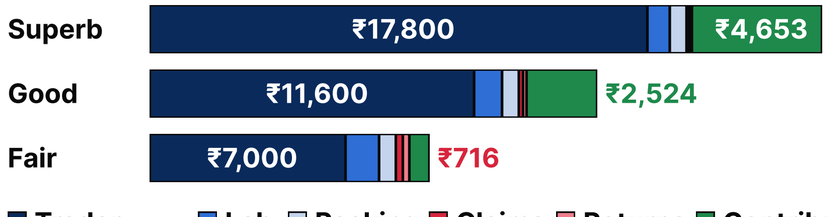
Test 4 requires all Exhibit 5 controls; A4.

Sana asks	The obvious answer	What we found, and what we do	Proof	Result
1 What holds Tikau back	Demand: cut prices and buy ads	Trust behind a full lab: 7,200 sold of 7,200 sellable; 5 of 7 lost-sale reasons are trust	Ads lose ₹2.1 lakh a month	EBITDA −₹14.1 lakh a month
2 Who to win	The biggest segment, or value seekers	Family buyers on Good phones: no supply cap, ₹2,699 a phone after the plan	A Fair phone earns ₹164 after B2	Har phone ki kundli. Saal bhar ki zimmedari.
3 What, and when	Price push (A), or every option at once	B1 and B2 set up in January, third shift from 1 February; no A, B3 or C before April	5 of 84 clear the conditional screen	March EBITDA +₹5.6 lakh
4 How customers hear	Discount banners and countdown timers	The Tikau Report above the price on every page; stores, referrals, labelled creators	3 claims made, 2 refused	Proof required; A4
5 How we know	A monthly P&L review	Weekly orders, return rate and Good battery swaps, each with a trigger	Orders under 1,838 a week	April: trade in on new money

Source: case Exhibits 1 to 6 and page 7 | team model 9b3628d94d (T12 to T16) | every number is rebuilt on slides A1 and A2

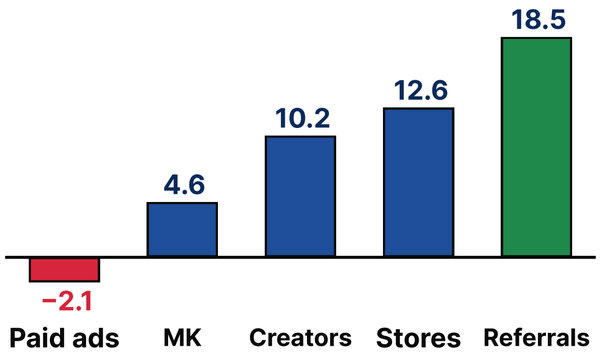
Every phone earns money before marketing, but Fair earns ₹716 to Superb's ₹4,653, ads lose ₹2.1 lakh a month and Tikau loses ₹14.1 lakh a month

ONE PHONE, ₹: WHERE THE SELLING PRICE GOES



Grade	Contribution	Share of price	Claims, returns
Superb	₹4,653	19.4%	₹147
Good	₹2,524	15.8%	₹276
Fair	₹716	7.2%	₹484

CONTRIBUTION AFTER MARKETING BY CHANNEL, ₹ LAKH A MONTH



Channel	Orders	Per order	Dashboard	True cost
Paid ads	2,600	−₹81	₹2,308	₹2,308
MegaKart	2,000	₹231	₹0	₹1,692
Creators	900	₹1,137	₹2,000	₹2,000
Stores	1,000	₹1,258	₹2,100	₹2,100
Referrals	700	₹2,648	₹350	₹350

MegaKart's ₹0 dashboard hides a 12% fee: ₹1,692/order; ₹231 remains. Working: A1.

THE MONTH TODAY, ₹ LAKH



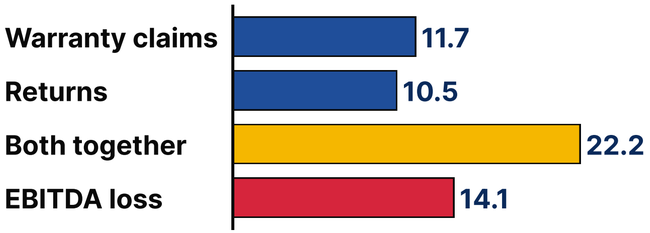
x 12 = −₹169.6 lakh, against the case's −₹170 lakh for the last twelve months.

RETURN RATE BY CHANNEL AND GRADE, %

	Superb	Good	Fair	Channel
Performance ads	4.0	7.0	13.0	8.8
MegaKart	4.0	7.0	13.0	9.4
Creators	6.0	10.5	19.5	9.6
Stores	2.0	3.5	6.5	3.1
Referrals	4.0	7.0	13.0	6.9

8.1% of phones come back within 7 days; the board's March limit is 6.0%

WHAT DISTRUST COSTS EACH MONTH, ₹ LAKH

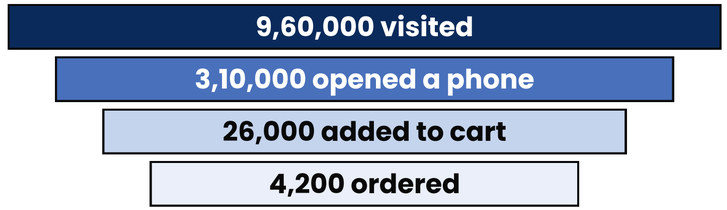


More than the loss: 468 claims and 582 returns a month cost ₹22.2 lakh. Repairs take 9 days and 38% of claims need a second visit.

Source: case Exhibits 2 and 3, page 3 | team model 9b3628d94d (T2 to T6) | dashboard cost per order as the growth team reports it: fixed spend plus coupons over orders

Trust and a full lab constrain growth: 91.6% leave without adding to cart; battery concern exceeds price concern

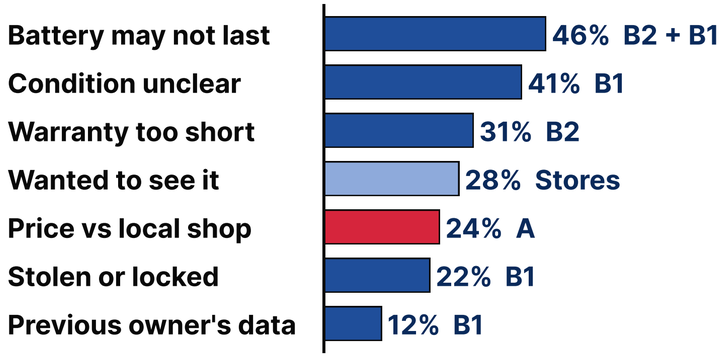
WEB FUNNEL; STORES / MK EXCLUDED



91.6% leave the phone page without adding to cart

Step	People	Conv.
Visited	9,60,000	
Opened a page	3,10,000	32.3%
Added to cart	26,000	8.4%
Ordered	4,200	16.2%
Store pickup	n/a	33%

WHY 1,500 VISITORS DID NOT BUY, %



Multi-response: shares overlap. Bar tags map concerns to options.

WHAT WOULD MAKE THEM BUY, %

	Share	Fixed by
85% battery promise	44%	B2
12 month warranty	39%	B2
Photos, battery, IMEI	36%	B1
Old phone in part	31%	C
A price 5% lower	22%	A

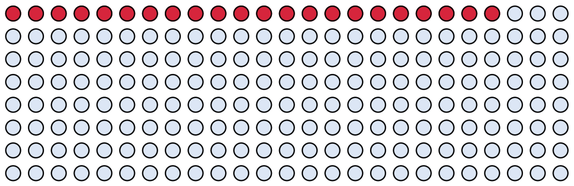
Stated interest: battery 85% 44%; 5% price cut 22%. A cuts Fair by 3%, a different offer.

DEMAND IS NOT THE LIMIT: THE LAB IS FULL



8,000 tested, 10% fail, 7,200 sellable, all sold. Any extra order needs a third shift; without one the case's rule drops it.

SPOT CHECK: 22 OF 200 GOOD



22/200 were below listed 80%. Scaling 11% to 3,190 Good gives ~351/month, indicative only (A14). Own-lab proof: A4.

FIVE WHYS TO ONE ROOT CAUSE

Why	Finding
1 Loss	claims and returns cost ₹22.2 lakh a month
2 Returns	buyers doubt battery (46%) and condition (41%)
3 Doubt	battery figures come from the trader's tool
4 Tool	the lab runs at 7,200 of 7,200, no spare time
5 Root	Tikau sells what it cannot yet prove

Source: case Exhibit 1 (1,500 visitors who left, October funnel), Exhibit 4 (lab, supply, November spot check), Exhibit 5 | team model 9b3628d94d (T7 to T9)

Family buyers fit the Good-phone promise: no Good supply cap, with contribution rising from ₹2,524 to ₹2,699

SEGMENT SCORECARD, 1 WEAK TO 5 STRONG

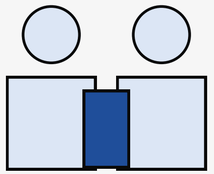
	Size	Margin	Headroom	Returns	Worry fit	Lever fit	Total
Flagship on a budget	5	5	1	5	3	3	22
Value seekers	4	1	5	1	1	1	13
Family buyers	3	3	5	3	5	5	24
Upgraders	2	4	3	4	2	1	16

Score = six equal 1-5 judgements; max 30. Sensitivity: A3.

Family 24; flagship 22. Double size or margin weight: tie 27 (T26).

Superb cap 2,000; ~70 orders rationed. No segment allocation assumed.

FAMILY BUYER: CASE + INFERENCE

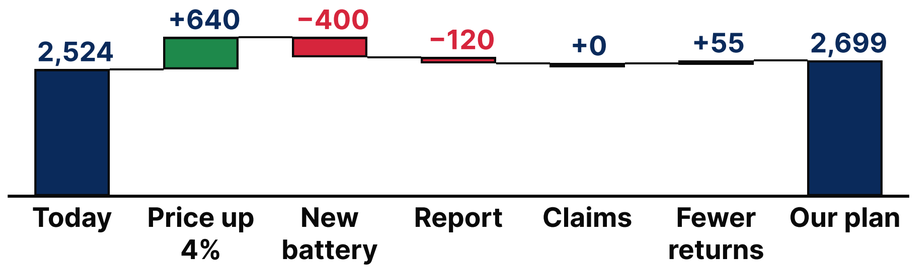


Illustration

Trait	Evidence
Who buys	21%, mostly Good phones
Wants	to know who will fix it
Visitor data	Battery will not last, 46%
Visitor data	Condition unclear, 41%
Channel plan	Stores and referrals: proposed

In a customer's words: "Phone theek hai, par bharosa nahi hua." The phone is fine, but I never came to trust it.

ONE GOOD PHONE, TODAY TO OUR PLAN, ₹



Fair goes the other way: ₹716 today, ₹164 after the battery and the report. We do not chase value seekers.

POSITIONING

For families buying a phone for a parent or a child, who need it to last and need to know who will fix it, Tikau sells refurbished phones that each come with a lab report, battery health of 85% or more and a 12 month warranty, because every phone is tested and its battery read in Tikau's own lab.

Proof 1: battery read in our lab

Proof 2: IMEI checked, data wiped

Proof 3: 12 months, repaired by Tikau

THE LINE A CUSTOMER REMEMBERS



Har phone ki kundli.
Saal bhar ki zimmedari.

Every phone's full report. A year of responsibility. A kundli is the chart a family reads before a big decision.

Source: case Exhibit 1 (segments, reasons, offers), Exhibits 2, 4 and 6 | team model 9b3628d94d (T10, T11) | scores are team judgement on the case data named in each row

THE OPTIONS

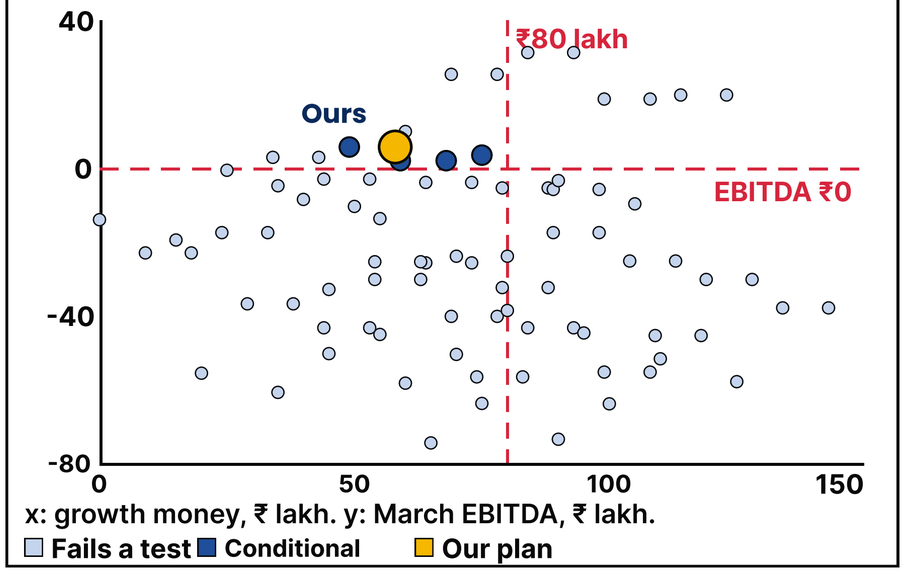
We ran all 84 plans the options allow: 5 pass the board's four tests, and only certify, battery and a February shift ends Q1 within ₹0.5 lakh of doing nothing

EACH CAMP'S PLAN AGAINST THE FOUR BOARD TESTS

	Growth money	Mar EBITDA	Q1 vs nothing	Mar returns	Test 1	Test 2	Test 3	Test 4
Do nothing	0.0	−14.1	0.0	8.1%	Pass	Fail	Fail	Fail
Rohan: price push (A)	20.0	−55.7	−83.2	8.9%	Pass	Fail	Fail	Fail
Rohan with a shift	38.0	−36.9	−45.6	8.9%	Pass	Fail	Fail	Fail
Kabir: B1 + B2, no shift	40.0	−8.6	−29.0	4.5%	Pass	Fail	Pass	Pass
Stores: trade in (C)	35.0	−4.9	−20.7	8.0%	Pass	Fail	Fail	Fail
B2 + shift, no report	43.0	2.8	+9.0	6.5%	Pass	Pass	Fail	Fail
All: B1, B2, C, shift	93.0	31.2	+5.9	4.5%	Fail	Pass	Pass	Pass
B1 + B2 + C, no shift	75.0	3.4	−44.4	4.5%	Pass	Pass	Pass	Pass
Ours: B1 + B2, Feb shift	58.0	5.6	−0.5	4.5%	Pass	Pass	Pass	Pass

Test 1: money up to ₹80 lakh. Test 2: March EBITDA ₹0 or more. Test 3: returns up to 6.0%. Test 4: Exhibit 5.

84 PLANS: MARCH EBITDA AND GROWTH MONEY



CONCLUSIVELY: THE OBVIOUS ANSWERS FAIL

Rejected plan	The number that kills it	Verdict
Price push (A)	Fair falls to ₹416; March −₹55.7 lakh	Fails tests 2 to 4
Certify, no shift	1,198 orders a month miss the lab	Fails test 2 (−₹8.6 lakh)
Trade in now (C)	needs ₹93.0 lakh with B1, B2, shift	Fails test 1

THE FIVE PLANS THAT PASS, ₹ LAKH

Plan	Growth money	Mar EBITDA	Q1 vs nothing	Mar returns
B1 + B2 + shift from Feb	58.0	5.6	−0.5	4.5%
B1 + B2 + shift from Mar	49.0	5.6	−14.8	4.5%
B1 + C + shift from Feb	68.0	1.9	−26.6	5.6%
B1 + C + shift from Mar	59.0	1.9	−34.4	5.6%
B1 + B2 + C	75.0	3.4	−44.4	4.5%

The next best plan ends Q1 ₹14.2 lakh lower.

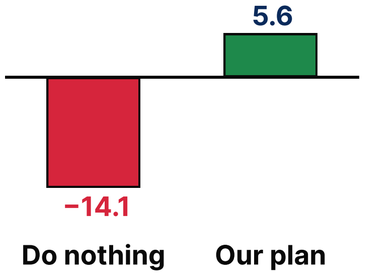
Source: case E6 and page 7 | model 9b3628d94d, T12-T14/T23 | Test 4 uses B1 as a screening proxy; Exhibit 5 controls remain mandatory (A4) | Q1 ranking: A3

Our plan uses ₹58.0 lakh and improves March EBITDA by ₹19.7 lakh to +₹5.6 lakh; the claims test remains conditional on Exhibit 5 controls

MARCH EBITDA, ₹ LAKH A MONTH

+19.7

₹ lakh a month better than doing nothing, from 1 February



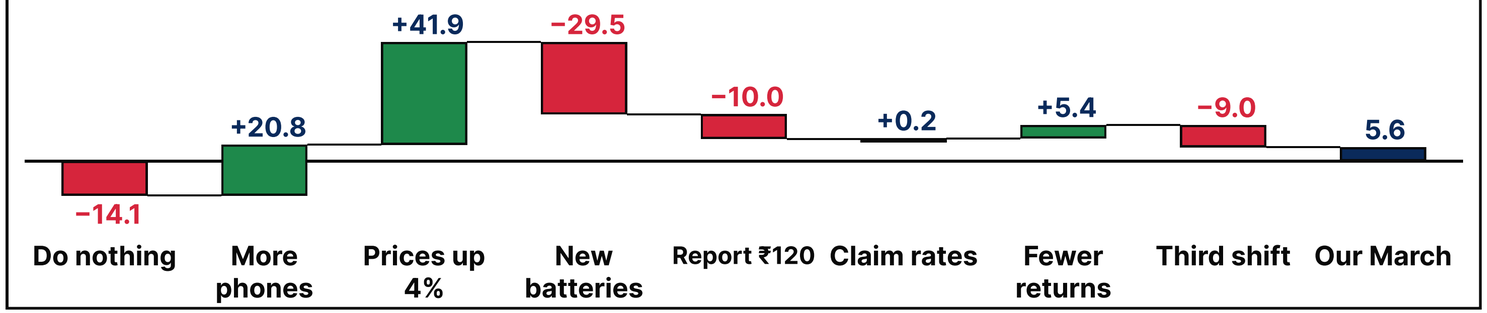
₹58.0 lakh	4.5%	8,328	93%
of ₹80 lakh used	returns; limit 6.0%	phones in March	of the 9,000 lab limit

THE DECISION TABLE (CASE PAGE 7)

Your plan	January	February	March
Options live	Setup paid	B1, B2, shift	B1, B2, shift
Phones sold	7,200	8,328	8,328
Contribution after marketing (₹ lakh)	43.9	72.6	72.6
Growth money spent (₹ lakh)	40.0	9.0	9.0
EBITDA (₹ lakh)	−54.1	5.6	5.6
Return rate	8.1%	4.5%	4.5%

₹58.0 lakh	Pass	+5.6	4.5%	Pass	-₹0.5 lakh
total growth money, Jan to Mar	Test 1: ₹80 lakh limit	Test 2: March EBITDA, ₹ lakh	Test 3: March returns, limit 6.0%	Test 4: controls (A4)	Q1 against doing nothing (−42.9 vs −42.4)

MARCH EBITDA, DOING NOTHING TO OUR PLAN, ₹ LAKH



WHAT TIKAU WILL NOT DO, AND WHY

Not this	Because
A price push	March −₹55.7 lakh; a Fair phone falls to ₹416
B3 campaign	₹15 lakh a month buys ₹3.9 lakh of contribution
Trade in now	with B1, B2 and the shift it needs ₹93.0 lakh
More ad money	ads already lose ₹2.1 lakh a month
Value seekers	a Fair phone earns ₹164 after B2

₹22 lakh remains unspent. The plan protects March profit; Q1 is ₹0.5 lakh below doing nothing.

Source: team model 9b3628d94d (T15, T16); decision table format from case page 7 | EBITDA = contribution after marketing − ₹58 lakh fixed costs − growth money that month

Launch the Tikau Report before price, with claims released only after proof

February: B1 + B2 go live after January setup. Existing channel budgets stay fixed; every listing requires its own proof.

WHAT A CUSTOMER SEES FIRST

Tikau product page, Good grade

Illustrative page | refurbished Good



TIKAU REPORT

- **Battery health:** 87%, read in Tikau's lab on 2 Feb 2027
- **Condition:** light marks on the back, all shown above
- **IMEI:** checked; not reported lost or stolen
- **Data:** wiped, certificate included
- **Warranty:** 12 months, repairs in Tikau's lab
- **Returns:** 7 days

₹16,640 12 month warranty included

ADD TO CART

Not on this page: countdown timers, "only 1 left" and struck-out prices.

THE MAIN IDEA

Har phone ki kundli. Saal bhar ki zimmedari.

Show the report before the price. Every channel carries the same report: the product page, the store counter, the creator's video and the referral link. In English: every phone's full report, a year of responsibility.

1 February: report live on every listing, MegaKart included

Week 1: stores read batteries in front of buyers

Week 2: creators open the report on camera, labelled as ads

CHANNELS IN FEBRUARY

Channel	Role in the launch	Per order now	Per order, Mar	Spend
Referrals	Family code shares the report; same ₹350 coupon	₹2,648	₹2,923	coupons only
Stores	Battery read at the counter; report in the box	₹1,258	₹1,881	₹21 lakh
Creators	Open the report on camera, labelled as an ad	₹1,137	₹1,769	₹18 lakh
Paid ads	Report-first creative; no new money	−₹81	₹231	₹60 lakh
MegaKart	Report and 12 months on listings; no ads	₹231	₹92	12% fee

THREE PROPOSED CLAIMS AND TWO REFUSALS | RELEASE ONLY AFTER PROOF

	Required on each	Says refurbished	Allowed words	Proof to record	Verdict
Battery 85% or more, read in our own lab	Yes	Yes	Yes	Yes	MAKE
12 month warranty on every phone	Yes	Yes	Yes	Yes	MAKE
IMEI checked and data wiped, with a certificate	Yes	Yes	Yes	Yes	MAKE
"As good as new, with genuine parts"	No	n/a	No	No	DO NOT
"Cheaper than PhoneBazaar for 12 months"	No	n/a	Yes	No	DO NOT

Source: case Exhibits 3, 5 and 6 | team model 9b3628d94d (T21) | the product page is an illustration: the battery reading shown is an example, not a measured phone

SIGNALS AND APRIL

Track deliveries, completed return cohorts and Good battery swaps; investigate warnings before March

1 ORDERS A WEEK

1,923

All-channel deliveries
Owner: Growth

2-week trigger: fewer than 1,838/week

EBITDA zero: ~1,838/week. Investigate early; A4.

Report demos; review delivery and channel gaps.

2 RETURN RATE

4.5%

Closed 7-day cohort
Owner: Service

Warning: cohort returns ≥5.5%

5.5% warns; 6.0% is the board limit, not profit zero.

Audit closed cohorts; retest affected batches.

3 GOOD PHONES NEEDING A BATTERY

40%

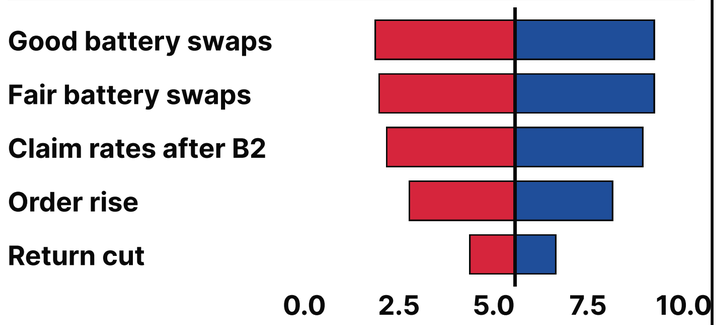
Good swaps / tested
Owner: Lab

Warning: Good swaps ≥50%

At 50%: +₹1.9 lakh; EBITDA zero ~55%.

Retest batch; review mix; rerun forecast.

WHAT MOVES MARCH EBITDA, ₹ LAKH



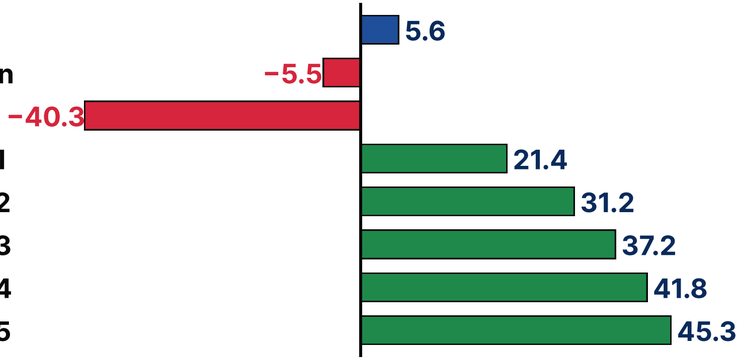
Red: the input 25% worse. Blue: 25% better. No single input takes March below zero; all five worse together give −8.7 ₹ lakh.

THREE SCENARIOS, ₹ LAKH

	Bear	Base	Bull
Feb EBITDA	−8.7	5.6	20.5
Mar EBITDA	−8.7	5.6	20.5
Q1 vs nothing	−29.2	−0.5	+29.2
Mar returns	5.3%	4.5%	3.8%
All tests	Fail	Pass	Pass

Bear: every uncertain effect 25% worse at once. The weekly numbers show it by the second week of February.

APRIL OPERATING EBITDA, BEFORE SETUP



C: ₹35 lakh setup; ~1.75-month incremental recovery. April −₹13.6 lakh after setup.

Source: team model 9b3628d94d (T17 to T20) | weekly = monthly / 4.33 (A13) | stress tests move uncertain effects 25% (A16) | April assumes new growth money (A15)

A1. Every number on the main slides rebuilds from the case exhibits and twenty stated assumptions, and the model passes 13 of 13 checks

CONTRIBUTION PER PHONE, ₹ (EXHIBIT 2)

Line	Superb	Good	Fair
Average selling price	₹24,000	₹16,000	₹10,000
Price paid to the trader	−₹17,800	−₹11,600	−₹7,000
Testing and repair	−₹800	−₹1,000	−₹1,200
Packing, delivery, payment	−₹600	−₹600	−₹600
Claims: rate x ₹2,500	−₹75	−₹150	−₹250
Returns: rate x ₹1,800	−₹72	−₹126	−₹234
Contribution per phone	₹4,653	₹2,524	₹716

Formula: contribution = price − trader price − lab − ₹600 packing − claim rate x ₹2,500 − return rate x ₹1,800. A return does not reverse the sale (Exhibit 2).

CHANNEL WORKING, ₹ LAKH A MONTH (EXHIBIT 3)

Channel	Orders	Phones	Fee	Fixed	Coupons	After mkt	Returns
Paid ads	2,600	57.9	0.0	−60.0	0.0	−2.1	8.8%
MegaKart	2,000	38.5	−33.8	0.0	0.0	4.6	9.4%
Creators	900	28.2	0.0	−18.0	0.0	10.2	9.6%
Stores	1,000	33.6	0.0	−21.0	0.0	12.6	3.1%
Referrals	700	21.0	0.0	0.0	−2.4	18.5	6.9%
Total	7,200	179.2	−33.8	−99.0	−2.4	43.9	8.1%

Rule: each channel's phones use its own return rate (creators 1.5x, stores 0.5x), then lose fixed spend, MegaKart's 12% fee and ₹350 referral coupons. EBITDA = 43.9 − 58.0 = −14.1.

Cost layers: phone contribution → channel costs → ₹58 lakh company fixed costs → monthly growth money.

ASSUMPTIONS WE STATED, A01 TO A20 (EACH ALSO SITS ON THE SLIDE THAT USES IT)

ID	Assumption
A01	Lever A's ₹10 lakh counts once, as growth money, not in ads' fixed spend
A02	Every channel's fixed spend stays as today in every plan and month
A03	Commission, coupons and per-phone costs apply to delivered orders only
A04	Rationing covers trade in orders too: Superb first, then all orders pro rata
A05	Return cuts from B1 and B2 multiply each channel's own return rate
A06	The 15% trade in failure rate applies equally to every grade
A07	30% of all trade ins, failed or passed, also buy a Tikau phone
A08	From February Superb is graded on Tikau's reading; none changes grade
A09	January is today in every plan; doing nothing repeats today three times
A10	A third shift adds 1,800 sellable phones for ₹9 lakh of growth money a month

ID	Assumption
A11	B2's 4% price rise also raises MegaKart's 12% commission
A12	The ₹120 report and B2 battery costs apply to trade in buyers' phones too
A13	Weekly figures use 4.33 weeks a month
A14	The spot check rate (22 of 200) applies to every Good phone sold
A15	April brings new growth money; trade in ramps 1,000, 1,800, then +800 a month
A16	Stress tests move only what Tikau does not control, 25% worse or better
A17	The case's ₹139 crore is read as selling price before commission
A18	August to October averages stand for November to January
A19	Store, creator and MegaKart return multipliers hold under B1 and B2
A20	Only traded phones that pass save 8%; failed ones go for parts at cost

Source: case Exhibits 2 to 4, page 3 | team model 9b3628d94d (T1 to T6) | rerun: python -X utf8 marksoc-case/model/dobaara_model.py

A2. Orders rise 16.6% and returns fall to 56% of today's rate under B1 and B2, the lab caps at 9,000, and March stays above zero down to a 10.5% order rise

HOW THE OPTIONS COMBINE (EXHIBIT 6 RULES)

Effect	Working
Orders	1.08 × 1.08 = 1.1664, so +16.6% in every channel
Returns	0.70 × 0.80 = 0.56 of each channel's rate; March 4.55%
Superb rationing	demand 2,070 against 2,000: 70 orders (3.4%) lost
Lab	sold 8,328 of 9,000 with the third shift (93%)
Prices	Superb ₹24,960, Good ₹16,640, Fair ₹10,000
Batteries	Good 40% x ₹1,000 = ₹400; Fair 70% x ₹800 = ₹560
Claims	Superb 4%, Good 6%, Fair 9% at ₹2,500
Growth money	Jan ₹40.0 (B1 ₹15, B2 ₹25); Feb ₹9.0; Mar ₹9.0 lakh

CONTRIBUTION PER PHONE UNDER EACH OPTION, ₹

Option	Superb	Good	Fair
Today	₹4,653	₹2,524	₹716
A	₹4,653	₹2,524	₹416
B1	₹4,555	₹2,442	₹666
B2	₹5,602	₹2,789	₹228
B1 + B2	₹5,500	₹2,699	₹164
A + B1 + B2	₹5,500	₹2,699	−₹136

Read: B1 and B2 together add ₹847 to a Superb phone and ₹175 to a Good one, but cut a Fair phone to ₹164. With A on top a Fair phone loses money.

CHANNELS IN MARCH UNDER OUR PLAN

Channel	Phones	After mkt, ₹ lakh	Per order	Returns
Paid ads	3,012	7.0	₹231	4.9%
MegaKart	2,325	2.1	₹92	5.3%
Creators	1,035	18.3	₹1,769	5.4%
Stores	1,149	21.6	₹1,881	1.8%
Referrals	807	23.6	₹2,923	3.9%

MegaKart earns ₹92 an order in March: its 12% fee grows with the 4% price rise and its mix leans to Fair. We add no MegaKart spend.

MARCH BOUNDARIES

Input	Plan	Boundary
Order rise from B1, B2	+16.6%	+10.5%
Orders a week	1,923	1,838
Good battery swaps	40%	55%
Fair battery swaps	70%	97%
Returns kept	56%	74%

Returns: 74% remaining reaches the 6% board limit; not EBITDA zero.

One input changes; all others fixed. Joint bear: −₹8.7 lakh (Signals).

SHORT FORMS

Short	Meaning
A to C	the case's options
IMEI	a phone's identity number
Q1	January to March 2027
After mkt	contribution after marketing
Fee	MegaKart's 12% commission
₹ lakh	₹1,00,000
x	multiplied by

Source: case Exhibit 6 rules | team model 9b3628d94d (T11, T13, T17, T21) | short forms used in the deck sit bottom right

A1 | A2 | A3 | A4

APPENDIX 3 | MODEL DEFENCE

February capacity gives the strongest feasible Q1 result

Case-only model | Money in ₹ lakh | March volumes after sequential rationing

Feasible plans | conditional on claims controls

Plan / shift	Growth	Mar E	Q1 E	Returns
B1+B2 / Feb	58	+5.6	−42.9	4.5%
B1+B2 / Mar	49	+5.6	−57.2	4.5%
B1+C / Feb	68	+1.9	−69.0	5.6%
B1+C / Mar	59	+1.9	−76.8	5.6%
B1+B2+C / none	75	+3.4	−86.8	4.5%

E = EBITDA; all five pass the numerical screen.

Two-input sensitivity | March EBITDA, ₹ lakh

Rows: combined uplift | Columns: Good battery swaps

Uplift / swap	40%	50%	55.06%	60%
8%	−3.8	−7.2	−9.0	−10.7
10.51%	0.0	−3.5	−5.3	−7.1
16.64%	+5.6	+1.9	0.0	−1.8
20%	+7.6	+3.8	+1.8	−0.1

Gold = zero boundary; blue = profit; red = loss.

Ordered rationing | selected March plans

Plan / shift	Demand	S factor	Lab	Sold
B1+B2 / Feb	8,398	96.6%	100.0%	8,328
B1+B2 / none	8,398	96.6%	86.5%	7,200
A / none	9,500	100.0%	75.8%	7,200
B1+B2+C / Feb	8,938	100.0%	100.0%	8,938

Demand → Superb supply → total lab → delivered phones

Factors retain orders proportionately; no demand carries over.

Third-shift timing | B1+B2 throughout

Shift	Feb E	Mar E	Q1 E
Feb	+5.6	+5.6	−42.9
Mar	−8.6	+5.6	−57.2
None	−8.6	−8.6	−71.4

Team objective: highest Q1 EBITDA among feasible plans.

Rank on unrounded totals; display money to ₹0.1 lakh.

Phone totals are rounded expectations; factors shown as %.

Source: Case E3/E4b/E5/E6 | T22-T25 | Independent audit | Model 9b3628d94d

A4 | OPERATING & CLAIMS CONTROLS

Three claims need proof; weekly warnings protect the plan

Proposed controls and owners. Compliance is conditional; no tests, approvals or service gains are claimed as completed.

Claims and proof | proposed owners and release stops

Claim / refusal	Evidence required	Owner	Stop / correction
Tikau Report	Own photos, lab reading, IMEI and wipe certificate	Lab + product	Hold phone if record is missing or mismatched
Battery ≥85%	Own-lab release reading; replacement record	Lab	Hold below threshold; no runtime guarantee
12-month warranty	Approved terms and accessible claim route	Service	Hold unsupported term; no instant-repair claim
REFUSE: Like new	Refurbished wording required on every listing	Commerce	Remove prohibited condition wording
REFUSE: Genuine battery	Replacements are compatible, not brand-made	Growth	Use compatible wording; reject brand-parts claim

Also: comparisons, true prices/stock, verified reviews and ad labels. Full matrix: research §28.

Weekly definitions | proposed conventions

Deliveries ~1,923/wk; zero ~1,838
All-channel deliveries; month ÷ 4.33 weeks
Growth + Ops: 2 weeks below ~1,838, reforecast

Returns ~4.5%; warn 5.5%
7-day returns / same mature cohort deliveries
Service lead: exclude immature; board max 6.0%

Good swaps 40%; warn 50%; EBITDA zero ~55%
Good needing swap / Good tested in same batch
Lab lead: audit lots; rerun delivered-cost model

Zero points hold other base inputs fixed. T27 / A13.

Service-control sequence | proposed, with phone-level release gates

1 Intake
Link identity + IMEI; bar lost/stolen units

2 Measurement
Own-lab battery test; replace if required

3 Report / listing
Link photos + wipe; match claim to device

4 Sale
Say refurbished; show terms + claim route

5 Repair
Record claim stage, age and repeat visit

6 Feedback
Trace issue to cohort; correct cause + copy

Missing proof, failed wipe or barred IMEI: stop the sale. No new repair-time target.
Tested Good share must represent delivered Good cost; reconcile held versus delivered phones.

Risk → paired response | proposed

Shift delay → rerun start timing [T25]
Good swap cost → audit lot + reforecast [T27]
Weak return cut → diagnose mature cohorts [T27]
Lab proof mismatch → hold unit claim [E5a]
Wipe / IMEI failure → stop affected sale [E5f]
Repair backlog → inspect stages + repeat visits [E4d]

Case: E5a-f; E6B1/B2/SH; D2; F16.
Model: T24/25/27. Research: §§28/29/32.
Full 12-risk register: §32; no observed outcomes.

THANK YOU

THEREFORE

Certify every phone, promise 85% battery and 12 months, open the third lab shift on 1 February and target family buyers. March EBITDA reaches +₹5.6 lakh on ₹58.0 lakh growth money. Numerical tests pass; claims require Exhibit 5 controls.

March EBITDA +₹5.6 lakh	Return rate 4.5%, limit 6.0%	₹58.0 of ₹80 lakh spent

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member
Dayita Arora	Ramjas College, University of Delhi	Member

Audited model and defensive working: A1-A4. Supporting research is a separate deliverable.

harshkawatra11@gmail.com | Team IdeaForge | MarKSoc DTU Case Competition 2026, Project DOBAARA

Source: team model 9b3628d94d | case Exhibits 1 to 6